

Advanced (Habanero)

- Q1.** Which of the following numbers is divisible by 3 and 5?
(A) 15
(B) 20
(C) 25
(D) 30
(E) 35
- Q2.** If a divides b and c divides d , what can be concluded about ac and bd ?
(A) ac divides bd
(B) bd divides ac
(C) ac and bd are prime
(D) ac and bd are relatively prime
(E) None of the above
- Q3.** What is the remainder when 17 is divided by 5?
(A) 1
(B) 2
(C) 3
(D) 4
(E) 0
- Q4.** Which number is divisible by 9?
(A) 18
(B) 27
(C) 36
(D) 45
(E) 63
- Q5.** Perform the division: $48 \div 6$
(A) 6
(B) 7
(C) 8
(D) 9
(E) 10
- Q6.** What is the quotient when 99 is divided by 11?
(A) 8
(B) 9
(C) 10
(D) 11
(E) 12
- Q7.** If 24 divided by 3 is 8, then 36 divided by 3 is
(A) 10
(B) 11
(C) 12
(D) 13
(E) 14
- Q8.** A number is divisible by 10 if and only if it is divisible by 2 and
(A) 3
(B) 5
(C) 7
(D) 9
(E) 11

Open Ended Questions (FRQ)

Q9. Perform the long division: $413 \div 17$. Express the result with remainder.

Q10. Find all the positive integers x for which 12 divides $x^2 - 5x + 6$. Justify your answer.

Chef's Tips

- Read each question carefully before answering
- Use a pencil to fill in the answer bubbles
- Check your work before submitting the test